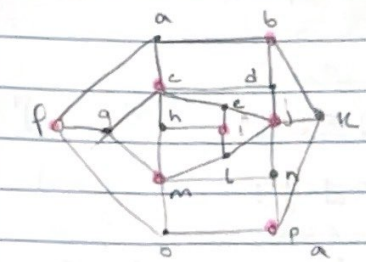
HW 8 due 3/8/24

1) 4)

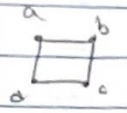


bipartite

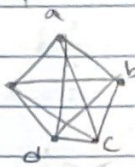
9. 7.

9 ≠ 7 so not Hamiltonian cycle

2) 10)



11) e



H: a-b-c-d-e (just outside edges)

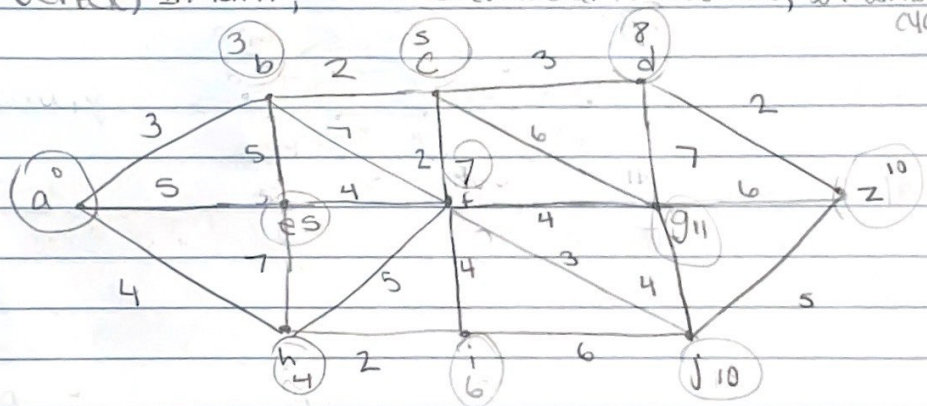
E: a-b-c-d-e-a-d-b-e-c-a

3) 14) If $n \geq 3$, K_n has a Hamiltonian cycle

Base: $n = 3$ △ has H cycle

Each K_n has a cycle of n vertices as a subgraph of the graph. Thus, ∃ a Hamiltonian cycle, because all the vertices are hit, and it is a cycle (same start+end vertex) In K_{n+1}, n+1 vertex connected to all others, so Hamiltonian cycle still exists.

4)



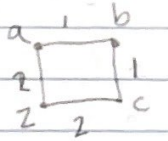
2) d(a,g) = 11

a-b-c-g

3) d(a,z) = 10

a-b-c-d-z

5) 10) False, alg fails for d(a,z):



alg would say shortest length is 4, but it's actually 2